

Inner Spring - Product Requirements Document

1. Executive Summary

Product Name: Inner Spring

Platform: Android Mobile Application

Target Audience: College Students

Mission: To provide stigma-free, accessible mental health support for college students, addressing anxiety, depression, burnout, sleep disorders, academic stress, and social isolation.

2. Product Overview

2.1 Product Vision

Inner Spring aims to be the go-to mental health companion for college students, offering comprehensive support through AI-guided assistance, professional connections, educational resources, and peer support in a safe, confidential environment.

2.2 Key Value Propositions

- **Stigma-Free Environment:** Anonymous and confidential platform
- **24/7 Accessibility:** Always-available AI support and resources
- **Comprehensive Care:** From self-help to professional referrals
- **Peer Community:** Moderated student support network
- **Data-Driven Insights:** Analytics for institutional mental health planning

3. Target Users

3.1 Primary Users

- **College/University Students in Jammu & Kashmir (Ages 18-25)**
 - Experiencing mental health challenges including academic stress, social isolation
 - Seeking confidential, culturally sensitive support
 - Need flexible, mobile-accessible resources in regional languages
 - May face connectivity challenges in remote areas

3.2 Secondary Users

- **Campus Counselors:** Professional mental health providers within university systems
- **Student Support Groups:** Peer-to-peer support networks and trained student volunteers

- **University Counseling Centers:** Administrative staff and mental health coordinators
- **Crisis Helpline Operators:** External mental health emergency services

4. Core Features & Requirements

4.1 AI-Guided First-Aid Support

Description: Interactive chatbot powered by Gemini API providing immediate mental health support with integrated psychological screening.

Functional Requirements:

- Natural language processing for student queries in Hindi, Urdu, Kashmiri, and English
- **Integrated Psychological Screening Tools:**
 - **PHQ-9 (Patient Health Questionnaire-9):** Depression severity assessment
 - **GAD-7 (Generalized Anxiety Disorder-7):** Anxiety severity measurement
 - **GHQ-12 (General Health Questionnaire-12):** General mental distress screening
- **Evidence-Based Categorization System:**
 - Automated scoring and classification (Mild, Moderate, Severe)
 - Standardized interpretation of screening results
 - Progress tracking through repeated assessments
- **Recommendation Engine Based on Screening Results:**
 - **Mild Symptoms:** Self-care resources, coping strategies, wellness content
 - **Moderate Symptoms:** Peer support referrals, structured self-help programs
 - **Severe Symptoms:** Professional counselor referrals, crisis intervention protocols
- Contextual coping strategy recommendations tailored to regional cultural context
- Crisis detection with automatic helpline number provision and escalation protocols
- Integration with existing campus counseling systems
- Conversation history for continuity of care
- Emergency helpline integration for immediate crisis intervention

Technical Requirements:

- Gemini REST API integration with psychological screening algorithms
- Retrofit for API communication with screening result processing
- Real-time response processing with clinical scoring calculations

- Secure storage of screening results and scores in Firestore
- Automated recommendation system based on standardized thresholds
- Progress tracking database for repeated assessments
- Secure data transmission compliant with DPDP Act 2023

4.2 Confidential Booking System

Description: Secure appointment scheduling with campus counselors and mental health services.

Functional Requirements:

- Anonymous user profiles with university verification
- Integration with existing campus counseling appointment systems
- Real-time availability checking for campus counselors
- Appointment confirmation and SMS/push notifications
- Cancellation and rescheduling with automated waitlist management
- Emergency appointment flagging for priority scheduling
- Automatic helpline number display for crisis situations

Technical Requirements:

- Firebase Realtime Database for instant messaging and peer interactions
- Firebase Cloud Messaging (FCM) for appointment notifications
- Google Calendar API integration for counselor availability sync
- Encrypted communication channels with end-to-end security
- CalendarView integration for intuitive date selection
- JobScheduler for automated reminder notifications

4.3 Psychoeducational Resource Hub

Description: Comprehensive library of mental health resources in multiple formats and languages.

Functional Requirements:

- Video content library with cultural context for J&K students
- Audio resources including guided meditation and relaxation in Hindi, Urdu, and Kashmiri
- Mental wellness guides addressing regional stressors (academic pressure, social expectations)
- Offline-first approach for areas with limited connectivity

- Content categorization by mental health topics and academic stress management
- Progress tracking with milestone achievements
- Resource recommendation engine based on user preferences

Technical Requirements:

- Firebase Cloud Storage with Glide for optimized media loading
- WorkManager for offline content synchronization
- SQLite/Room Database for offline storage with intelligent caching
- MPAndroidChart integration for progress visualization
- Material Design Components for consistent, accessible UI
- Content categorization system with search optimization

4.4 Peer Support Platform

Description: Student-led support groups and moderated forum for peer-to-peer mental health support.

Functional Requirements:

- Anonymous posting with verified student status
- Topic-based discussion groups (academic stress, social anxiety, homesickness)
- Student support group leadership and coordination tools
- Peer mentorship matching system
- Content moderation by trained student volunteers
- Crisis escalation protocols within peer discussions
- Support group scheduling and virtual meeting integration
- Cultural sensitivity guidelines and enforcement

Technical Requirements:

- Firebase Realtime Database for instant messaging infrastructure
- Google Cloud Natural Language API for automated content moderation
- Firebase ML Kit for inappropriate content detection
- Content flagging system with real-time processing
- User reputation system with Firestore backend
- FCM for real-time notifications and updates

4.6 Psychological Screening & Assessment Module

Description: Standardized mental health screening tools for evidence-based assessment and care coordination.

Functional Requirements:

- **PHQ-9 Implementation:**
 - 9-question depression screening questionnaire
 - Automated scoring (0-27 scale)
 - Classification: Minimal (0-4), Mild (5-9), Moderate (10-14), Moderately Severe (15-19), Severe (20-27)
 - Bi-weekly screening recommendations for monitoring
- **GAD-7 Implementation:**
 - 7-question anxiety disorder screening
 - Automated scoring (0-21 scale)
 - Classification: Minimal (0-4), Mild (5-9), Moderate (10-14), Severe (15-21)
 - Integration with panic disorder and social anxiety assessments
- **GHQ-12 Implementation:**
 - 12-question general mental health screening
 - Automated scoring with threshold-based interpretation
 - General psychological distress identification
 - Screening for overall mental wellbeing
- **Assessment Workflow:**
 - Initial comprehensive screening during onboarding
 - Periodic re-assessment scheduling based on risk levels
 - Progress tracking with visual analytics
 - Longitudinal data analysis for trend identification
- **Clinical Decision Support:**
 - Automated referral recommendations based on severity scores
 - Integration with appointment booking for high-risk scores
 - Crisis intervention triggers for severe classifications
 - Personalized resource recommendations aligned with screening results

Technical Requirements:

- Validated screening tool implementation with clinical accuracy
- Secure storage in encrypted Firestore collections with ProGuard protection
- Real-time scoring algorithms with immediate result processing
- MPAndroidChart integration for progress visualization and trend analysis
- WorkManager for scheduled follow-up screening reminders
- Integration with recommendation engine and referral system
- Compliance with psychological testing standards and ethics

4.7 Admin Dashboard

Description: Analytics platform for campus authorities to monitor mental health trends and plan interventions.

Functional Requirements:

- Anonymous usage analytics aggregated across J&K universities
- **Psychological Screening Analytics:**
 - Aggregated PHQ-9, GAD-7, and GHQ-12 score distributions
 - Trend analysis of severity classifications over time
 - Early warning indicators based on screening pattern changes
- Mental health trend identification specific to regional challenges
- Crisis intervention pattern analysis and early warning systems
- Resource utilization metrics by language and content type
- Academic calendar correlation with stress level patterns and screening scores
- Intervention effectiveness tracking for university counseling centers
- Anonymized demographic insights without personal identification
- Custom report generation for university administration

Technical Requirements:

- Firebase Analytics integration with custom screening metrics
- MPAndroidChart for data visualization and trend analysis
- Firebase Crashlytics for stability monitoring
- Firebase Performance Monitoring for optimization insights

- Role-based access control with enhanced security
- Real-time dashboard updates using Firebase Realtime Database
- DPDP Act 2023 compliant data handling with ProGuard protection

5. User Experience Requirements

5.1 Design Principles

- **Calming Aesthetic:** Soft, calm green color palette
- **Modern UI:** Clean, intuitive interface design
- **Accessibility:** WCAG 2.1 compliance
- **Privacy-First:** Clear privacy indicators and controls

5.2 Key User Flows

1. First-Time User Onboarding with Initial Screening
2. Psychological Screening Assessment (PHQ-9, GAD-7, GHQ-12)
3. AI Chat Interaction with Screening-Based Recommendations
4. Appointment Booking Process (Triggered by Severe Screening Results)
5. Resource Discovery Based on Assessment Results
6. Peer Forum Participation with Severity-Appropriate Matching
7. Crisis Escalation Pathway with Integrated Screening Data
8. Follow-up Screening and Progress Monitoring

6. Technical Architecture

6.1 Technology Stack

Core Development

- **Frontend:** Java, XML (Android Native)
- **Authentication:** Firebase Authentication
- **Database:** Firestore (cloud), SQLite/Room Database (local)
- **Storage:** Firebase Cloud Storage
- **Hosting:** Firebase Hosting
- **API Communication:** Retrofit

AI & Clinical Features

- **AI Integration:** Gemini REST API with clinical decision support
- **Screening Tools:** Validated PHQ-9, GAD-7, GHQ-12 implementations
- **Clinical Algorithms:** Standardized scoring and classification systems

Real-Time Features

- **Push Notifications:** Firebase Cloud Messaging (FCM)
- **Real-Time Chat:** Firebase Realtime Database
- **Live Updates:** Real-time synchronization for appointments and peer support

UI/UX Enhancements

- **Design System:** Material Design Components
- **Animations:** Lottie Android for smooth micro-interactions
- **Data Visualization:** MPAndroidChart for screening progress and analytics
- **Image Loading:** Glide for efficient image handling in forums and resources

Background Processing

- **Offline Synchronization:** WorkManager for background data sync
- **Scheduled Tasks:** JobScheduler for appointment reminders and screening follow-ups
- **Background Analytics:** Automated usage tracking and screening data processing

Security & Monitoring

- **Crash Reporting:** Firebase Crashlytics for stability monitoring
- **Performance Tracking:** Firebase Performance Monitoring
- **Code Protection:** ProGuard/R8 for code obfuscation and app security
- **Analytics:** Firebase Analytics with psychological screening metrics

Content Moderation & Safety

- **AI Content Filtering:** Google Cloud Natural Language API for peer forum moderation
- **On-Device ML:** Firebase ML Kit for inappropriate content detection
- **Automated Moderation:** Real-time content analysis for safety

Calendar & Scheduling Integration

- **External Calendar Sync:** Google Calendar API for counselor availability

- **Native Calendar UI:** CalendarView (Android native) for appointment booking
- **Schedule Management:** Integration with existing campus counseling systems

6.2 System Architecture

- **Client-Server Architecture** with enhanced real-time capabilities
- **Microservices approach** for scalability and feature modularity
- **Real-time synchronization** using Firebase Realtime Database for critical features
- **Background processing** with WorkManager for offline-first functionality
- **Secure API endpoints** with authentication and content moderation
- **MVVM Architecture Pattern** for maintainable code organization
- **Offline-first design** with intelligent sync capabilities
- **Progressive loading** with Lottie animations for enhanced UX

7. Security & Privacy Requirements

7.1 Data Protection & Compliance

- **DPDP Act 2023 Compliance:** Full adherence to India's Digital Personal Data Protection Act
- **Sensitive Personal Data Handling:** Special protocols for mental health conversations and student records
- **HIPAA-Inspired Best Practices:** Healthcare-grade security for mental health information
- End-to-end encryption for all sensitive communications
- Anonymous user identification system with university verification
- Local data encryption using Android Keystore

7.2 Privacy Features

- **Explicit User Consent:** Clear, granular consent mechanisms for data collection
- **Data Minimization:** Collect only essential information for service delivery
- **Right to Deletion:** Complete data removal upon user request
- **Data Portability:** Export user data in standard formats
- Anonymous profile creation with optional verification
- Crisis intervention data retention policies with legal compliance
- Regular privacy impact assessments

8. Performance Requirements

8.1 Response Time & User Experience

- AI chat responses: < 3 seconds with Lottie loading animations
- App launch time: < 2 seconds with optimized Material Design components
- Resource loading: < 5 seconds with Glide image optimization
- Appointment booking: < 10 seconds with real-time calendar updates
- Peer forum message delivery: < 1 second via Firebase Realtime Database
- Screening result processing: < 2 seconds with immediate visualization

8.2 Scalability & Reliability

- Support for 10,000+ concurrent users with Firebase Realtime Database
- 99.9% uptime availability with Firebase Crashlytics monitoring
- Auto-scaling cloud infrastructure with performance monitoring
- Efficient offline-online synchronization using WorkManager
- Real-time content moderation processing without user-facing delays
- Background processing optimization for appointment reminders and screening follow-ups

9. Compliance & Legal Requirements

9.1 Indian Data Protection Compliance

- **DPDP Act 2023:** Complete compliance with India's digital privacy regulations
- **Sensitive Personal Data:** Special handling protocols for mental health data
- **Data Fiduciary Obligations:** Proper consent management and data processing transparency
- **Cross-Border Data Transfer:** Compliance with international data sharing restrictions

9.2 Healthcare & Educational Compliance

- **HIPAA-Inspired Practices:** International best practices for health information security
- **Mental Health Confidentiality:** Professional ethics alignment with Indian psychological practice standards
- **Crisis Intervention Protocols:** Mandatory reporting aligned with Indian healthcare regulations
- **University Integration:** Compliance with institutional policies across J&K universities

9.3 Regional Considerations

- **Cultural Sensitivity:** Content and intervention approaches culturally appropriate for J&K

- **Language Compliance:** Accurate translations maintaining clinical precision
- **Local Healthcare Integration:** Alignment with regional mental health service protocols

10. Success Metrics

10.1 User Engagement

- Daily active users with completed psychological screenings
- Session duration and frequency correlated with severity levels
- Feature adoption rates across different risk categories
- User retention (weekly/monthly) by screening-based user segments
- Screening completion rates and follow-up adherence

10.2 Health Outcomes

- **Clinical Effectiveness Metrics:**
 - PHQ-9, GAD-7, GHQ-12 score improvements over time
 - Severity level transitions (severe to moderate, moderate to mild)
 - Screening-to-intervention conversion rates by severity level
- Crisis intervention response time and screening correlation
- Professional referral completion rates by severity classification
- Resource utilization effectiveness based on screening recommendations
- Peer support engagement levels by mental health status

10.3 Technical Performance

- App crash rates (< 0.1%) monitored via Firebase Crashlytics
- API response times tracked with Firebase Performance Monitoring
- System uptime percentage with real-time monitoring
- Data synchronization success rates using WorkManager analytics
- Real-time message delivery rates via Firebase Realtime Database
- Content moderation accuracy and response time metrics
- Calendar integration success rates and appointment booking completion

11. Risk Assessment

11.1 Technical Risks

- **API Dependencies:** Gemini API rate limiting, Google Cloud Natural Language API costs
- **Real-time Performance:** Firebase Realtime Database scaling during peak usage
- **Data Synchronization:** Offline-online sync conflicts with WorkManager
- **Third-party Integrations:** Google Calendar API reliability, FCM delivery rates
- **Content Moderation:** AI moderation accuracy and false positive rates
- **App Performance:** Memory usage with multiple real-time connections
- **Security:** ProGuard obfuscation effectiveness against reverse engineering

11.2 Legal/Compliance Risks

- Health data privacy violations with sensitive screening results
- Crisis intervention liability with clinical assessment data
- Psychological screening tool licensing and usage compliance
- Professional liability for automated clinical recommendations
- Platform content moderation challenges
- Cross-institutional data sharing regulations with clinical data

11.3 User Safety Risks

- Crisis situation management in remote areas of J&K
- Peer support content quality with cultural sensitivity
- AI recommendation accuracy for regional mental health contexts
- Professional referral availability in underserved areas
- Helpline accessibility during network connectivity issues
- Student support group leadership training and oversight

12. Implementation Timeline

Phase 1 (Months 1-3): Core Development

- Basic app structure and authentication
- Psychological Screening Tools Implementation (PHQ-9, GAD-7, GHQ-12)
- Clinical scoring algorithms and classification system

- AI chat integration with screening-based recommendations
- Resource hub foundation

Phase 2 (Months 4-6): Extended Features

- **Screening-based recommendation engine**
- **Clinical decision support system integration**
- Peer support platform with severity-appropriate matching
- Booking system integration with automated referrals
- Admin dashboard with screening analytics

Phase 3 (Months 7-8): Testing & Refinement

- **Clinical validation of screening tools and algorithms**
- Beta testing with select user groups and mental health professionals
- Security audits and compliance verification
- Performance optimization for screening workflows

Phase 4 (Months 9-10): Launch Preparation

- **Training for counselors on screening result interpretation**
- Content creation aligned with screening-based recommendations
- Staff training and onboarding with clinical protocols
- Marketing and awareness campaigns emphasizing evidence-based approach

13. Future Considerations

13.1 Potential Enhancements

- iOS version development for broader accessibility
- Wearable device integration for stress monitoring
- Advanced AI personalization with cultural context
- Expansion to other Indian states and regions
- Integration with government mental health initiatives
- Telemedicine integration with regional healthcare providers
- Academic performance correlation analytics

13.2 Scalability Planning

- Multi-university deployment across India
- Regional language expansion (Punjabi, Dogri, Ladakhi)
- Professional network expansion in remote areas
- Content localization for diverse Indian student populations
- Integration with National Mental Health Programme initiatives
- Advanced crisis intervention with government helpline integration